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Paper Code : IT-605D

ARTIFICIAL INTELLIGENCE

Allotted : 3 Hours *Full Marks* : 70

The figures in the margin indicate full marks.

ndidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

Choose the correct alternatives for the following : $10 \times 1 = 10$

i) A logic is a

a) formal language b) natural language

c) both (a) and (b) d) none of these.

ii) The semantic is a story about

a) legal expression mean

b) a file

c) a language

d) a function of computer.

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iii) "Colourless green ideas sleep furiously" is

a) semantically well formed

b) syntactically well formed

c) both (a) and (b)

d) none of these.

iv) A proof is a way to test whether

a) KB is entails a sentence

b) KB is assumes a sentence

c) KB is interprets a sentence

d) KB is holds a sentence.

v) Unification makes two expressions match each other exactly

a) by finding multiplier

b) by finding substitution

c) by finding adder

d) none of these.

vi) Horn clause is a clause that has

a) at most two positive literal

b) at least two positive literal

c) at most one positive literal

d) at most two negative literal.

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vii) Back chaining is a generalized form of

a) searching b) reasoning

c) stochastic process d) modus ponens.

viii) Blind search is

a) informed b) optimal

c) uninformed d) all of these.

ix) 'Heuristics' the generally refers to

a) counter propagation b) backtracking

c) rule of thumb d) none of these.

x) Worst time complexity of DFS

a) b^{d+1} b) b^d

c) bd d) none of these.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. 3 × 5 = 15

2. Convert the following to clause form :

$((\exists x)[P(x)] \vee (\exists x)[Q(x)]) \rightarrow (\exists x)[P(x) \vee Q(x)]$

$(\forall x)[P(x) \rightarrow (\forall y)[(\forall z)[Q(x,y)] \rightarrow (\forall z)[R(y,x)]]]$

3. Convert the following formula into one that has neither an existential quantifier nor a Skolem function :

$(\forall x,y)\{(\exists z)[On(x,z) \wedge \text{Above}(y,z)] \rightarrow Above(x,y)\}$

$(\forall x,y)\{(\exists z)[On(x,z) \wedge \text{Above}(y,z)] \rightarrow Above(x,y)\}$

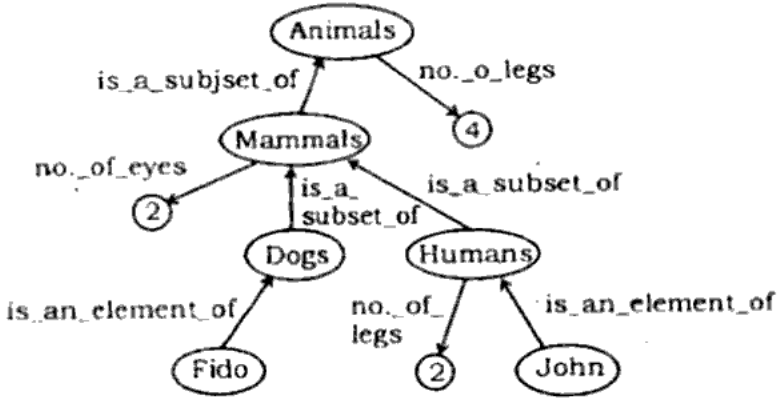
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4. Algorithm A^* does not terminate until a goal node is selected for expansion. However, a path to a goal node might be reached long before that node is selected for expansion. Why not terminate as soon as a goal node has been found ? Illustrate your answer with an example.

5. Use the min-conflicts method to solve the Four-Queens problem. Start with all of the queens on a main diagonal. Break ties randomly.

6. Express as predicate-calculus wffs the information in the semantic network shown here. Take care that the wffs are written in such a way that they explicitly state the inheritance cancellation information that is implicit in the inheritance hierarchy.



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GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. 3 × 15 = 45

7. The 'Missionary and Cannibals' problem has a famous history in AI. Here is one statement of the problem :

Three missionaries and three Cannibals come to a river. There is a boat on their side of the river that can be used by either one or two persons. How should they use this boat to cross the river in such a way that Cannibals never outnumber Missionaries on either side of the river ?

Specify the form of state descriptions, the starting state and the goal state for this problem.

Draw the entire state description graph and label its nodes. (You need to include only those states that are 'legal' — that is, states in which Cannibals do not outnumber missionaries on either side of the river.)

(3 + 3 + 3) + (3 + 3)

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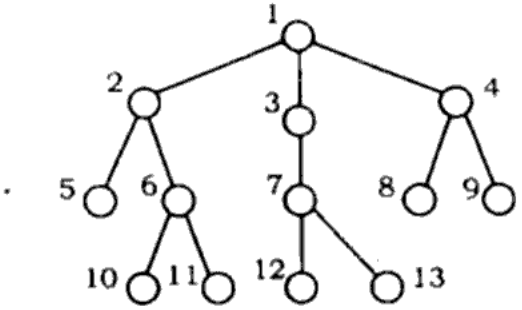
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8. List the order in which nodes are visited in the tree below for each of the following three search strategies (choosing leftmost branches first in all cases) :

a) Depth-first search

b) Depth-first iterative-deepening search (increasing the depth by 1 each iteration)

c) Breadth-first search.



5 + 5 + 5

9. An expert system is used to estimate whether or not people are good credit risks based on a number of factors that combine to produce a numeric score. To be judged a good credit risk, the score must be higher than some number, *N*. The designers of the expert system won't say what the value of *N* is, but we have learned that Pete is judged to be a good credit risk and that

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Joe's score is higher than Pete's. Prove that Joe would be judged to be a good credit risk also by using the following domain rules :

$\forall(x,y,z)[\text{Greater}(x,y) \wedge \text{Greater}(y,z) \rightarrow \text{Greater}(x,z)]$

$\forall(x)[C(x) \rightarrow \text{Greater}(\text{score}(x), N)]$

$\forall(x)[\text{Greater}(\text{score}(x), N) \rightarrow C(x)]$

And the following data :

$C(\text{Pete})$

$\text{Greater}(\text{score}(\text{Joe}), \text{score}(\text{Pete}))$

First, use these axioms and data to prove that Joe is a good credit risk, and then use explanation-based generalization techniques on your proof to establish the rule

$[\forall(x,y)\{[\text{Greater}(\text{score}(x), \text{score}(y)) \wedge C(y)] \rightarrow C(x)\}]$.

(3 + 3 + 3) + 6

10. a) Represent the following using predicate logic and draw the conclusion :
    i) Amal is an Indian
    ii) Bimal is an Indian
    iii) Amal is a leader
    iv) Every Indian is a man
    v) Every man is either likes a leader or dislikes a leader.

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b) What are the advantages of predicate logic over propositional logic ?

c) Prove that (i) Algorithm 'U' terminates successfully, (ii) If heuristic is consistent then that is admissible but the opposite is not true. 6 + 1 + (4 + 4)

11. Write short notes on any *three* of the following : 3 × 5
    a) Local maxima and global maxima
    b) State space representation
    c) Semantic net
    d) Dempster-Shafer theory
    e) Minimax Search Procedure.

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